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(Module 1)

Sr. no	Name Of Experiment	Page.no	Date	Signature
1.	**Creation of Test Suite Using Selenium IDE** Tool: Selenium IDE (Chrome/Firefox Extension) Install Selenium IDE and create a test suite containing a minimum of four test cases to validate different web page formats (HTML page, XML response, JSON API output, and a dynamic form page). Execute the suite and export the automation script in WebDriver format. .			
2.	**Cross-Website Functional Testing Using Selenium IDE** Tool: Selenium IDE Develop and execute a test suite for two different websites. Perform functional operations such as clicking links, filling forms, selecting dropdown options, and verifying displayed content. Analyze test results and generate execution logs.			
3.	**Manual Test Case Design and Execution** Tool: MS Excel / Google Sheets Prepare a Test Plan, Test Scenarios, and Test Cases for a given web application. Execute test cases manually and prepare a Test Execution Report and Defect Report as per STLC process.			
4.	**Selenium Server (Grid) Configuration and Remote Execution** Tool: Selenium Server (Grid) Install and configure Selenium Server in Hub-Node architecture. Execute a test script on multiple browsers using Remote WebDriver and compare cross-browser results			
5.	**Login Automation and Validation** Tool: Selenium WebDriver (Java/PHP) Write an automation script to perform login on a specified web page. Verify successful login using assertions such as URL validation, welcome message validation, and logout visibility check.			

6.	**Web Element Interaction and Synchronization Handling** Tool: Selenium WebDriver (Java) Develop a script to automate interaction with various web elements (textbox, radio button, checkbox, dropdown, alert, and frames). Implement implicit and explicit waits for synchronization handling.			
7.	**Data-Driven Testing Using Excel Integration** Tool: Selenium WebDriver + Apache POI Write a program to update 10 student records in an Excel file and validate changes on the web application. Perform data reading, writing, and verification using Apache POI library.			
8.	**Data Extraction and Score Analysis from Web Table** Tool: Selenium WebDriver Automate extraction of student marks from a dynamic web table and determine the number of students scoring above 60 in at least one subject. Compute total count and percentage.			
9.	**Object Identification and Counting on Web Page** Tool: Selenium WebDriver Write a program to identify and count the total number of objects present on a web page including links, images, buttons, and input elements. Display categorized counts.			
10.	**List and Combo Box Item Verification** Tool: Selenium WebDriver Automate the process of identifying a list or dropdown box on a web page and determine the total number of items. Validate default selected value and perform selection testing.			

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(Module 2)

Sr.No	Name Of Experiment	Page.no	Date	Signature
1.	**Checkbox Identification and Validation** Tool: Selenium WebDriver Write a program to count the total number of checkboxes on a web page, including			
2.	**Dynamic Web Table Handling and Sorting Validation** Tool: Selenium WebDriver Automate extraction of data from a dynamic web table. Validate sorting functionality and verify correctness of filtered results.			
3.	**File Upload and Download Automation** Tool: Selenium WebDriver + Robot Class Develop a script to automate file upload and download functionality. Validate successful upload and verify file existence after download.			
4.	**Screenshot Capture and Logging Mechanism** Tool: Selenium WebDriver + Log4j Implement screenshot capture on test failure and integrate logging mechanism using Log4j. Generate execution logs and analyze error messages**.**			
5.	**Load Testing Using Apache JMeter** Tool: Apache JMeter Design and execute load testing scenarios for a web application. Configure thread groups, ramp-up time, and loop count. Analyze response time, throughput, and error percentage using graphical reports.			
6.	**Bug Tracking and Defect Lifecycle Management** Tool: Bugzilla Log software defects using Bugzilla. Assign severity and priority levels. Track defect lifecycle stages and generate defect summary reports.			